

Time Series Forecasting
Stat 645
Drexel University
Project Guidelines

The objective of this project is to develop forecasting models for a time series dataset. Based on the characteristics of the data, you will be required to implement and test all of the relevant time series models that we learn about in class on the dataset that you select. You will be looking for the best model for forecasting and will compare each model to find it. You must do this as a group.

Data

With the exception of datasets associated with the text, you may choose any time series you like, but the instructor must approve all data. Send the data to me when you think that you've found the dataset you want to use. There are many time series datasets readily available on the internet. You could consider looking on google or Kaggle or the text's author's Time Series Data Library <http://robjhyndman.com/TSDL/>. In addition, there are 61,455 time series on the St. Louis Federal Reserve Website: <http://research.stlouisfed.org/fred/>. If you have a different dataset that you'd like to use feel free to ask me if it is acceptable.

Forecasting Models

In addition to using the tools that we learn in the course to explore/understand the time series characteristics of your data, you will be required to develop models for your time series using all of the models that we learn about in class.

Presentation

You will be submitting a video recording of your findings, along with a slide deck that goes along with your presentation. Your video must be at most seven minutes long (if you can do it in less time that is better). You will do best to write out exactly what you want to say, and stick to the script. Everyone in your group must participate in the presentation (do the talking for at least one slide), and you must include video of all speakers while speaking (can be in a window off to the side of the slide). You will lose points if you do not include video of speakers. You should include an introduction, conclusion and appropriate intermediate sections that tell your story and highlight what you did. Make sure to include any assumptions that you make, and include tables and charts that help tell your story. A good rule to follow is that if something doesn't contribute to telling the story in an efficient concise manner, then don't include it. You should be thinking about this in terms of Occam's Razor – the simpler the explanation the better, provided that the necessary information is communicated.

Please feel free to use me as a resource while working on the project. I'd be happy to answer any questions you have. The project is due on Tuesday of finals week.

You will also need to submit your R code and if you use one your .Rmd file.

Your analysis should include the following:

- A description of the data. Including time series plots.
- Alternative models that you considered.
- Judgment calls you made in choosing the final model.
- Graphs showing how well the model performs in forecasting.
- Measurement of forecast accuracy.

You must include a holdout sample of at least 10% of your data in order to test your model(s).

Grading

You will be graded on the *quality and clarity of your analysis* and on the *clarity of your presentation*. Along with your completion of the Final Project - Quizzes.

Please feel free to use me as a resource while working on the project. I'd be happy to answer any questions you have.

The project is due at 9 pm on Tuesday of finals week. Submit your presentation, .R or .Rmd files to bb learn.